INCH-POUND

MIL-DTL-17/176E
w/AMENDMENT 1
21 September 2023
SUPERSEDING
MIL-DTL-17/176E
19 June 2015

MILITARY SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE, TWIN, M17/176-00002 and M17/176-00003

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

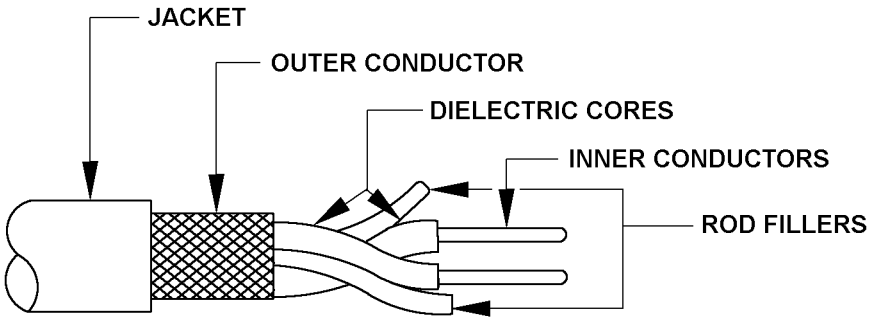


FIGURE 1. Configuration M17/176-00002 and M17/176-00003.

TABLE I. Description.

Components	Construction details
Inner conductors 1/	Two number 24 AWG conductors, each consisting of nineteen strands of number 36 AWG silver-coated, high strength copper alloy wire. Diameter of each strand: 0.0050 inch. Overall diameter of each conductor: .023 inch minimum, .024 inch maximum.
Dielectric cores (M17/176-00002)	Two PTFE cores twisted together with a left hand lay of 11 to 14 twists/foot. Fill to round with two PTFE rod fillers. Diameter of each dielectric core: .042 ± .002 inch. Core colors: One blue, one white.
Dielectric cores (M17/176-00003)	Two ECTFE or ETFE foam cores twisted together with a left hand lay of 11 to 14 twists/foot. Fill to round with two ECTFE or ETFE rod fillers. Diameter of each dielectric core: .042 ± .002 inch. Core colors: One blue, one white.
Outer conductor 1/	Single braid of AWG number 38, silver-coated, high strength copper alloy wire. Diameter: .102 inch maximum. Coverage: 93.3 percent nominal, 90 percent minimum. Carriers: 16 Ends: 6 Picks/inch: 11.2 ± 10 percent. Braid angle: 22 nominal.
Jacket (M17/176-00002)	Type XIII: PFA. Overall diameter: .129 ± .005 inch. Color: Translucent blue.
Jacket (M17/176-00003)	Type: PFA, FEP, ETFE or ECTFE. Overall diameter: .125 ± .005 inch. Color: Translucent blue, white, or clear.

1/ Silver-coated in accordance with ASTM B298, high strength copper alloy in accordance with ASTM B624.

AMSC N/A FSC 6145

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ENGINEERING INFORMATION:

Continuous working voltage: 750 V rms, maximum.

Operating frequency: 10 GHz, maximum.

Velocity of propagation: 68 percent nominal for M17/176-00002.
72 percent nominal for M17/176-00003.

Operating temperature range: -55° to +200°C for M17/176-00002.
-55° to +150°C for M17/176-00003.

Inner conductor properties:

DC resistance (maximum at 20°C): 2.84 ohms per 100 feet.

Elongation: 6 percent, minimum.

Tensile strength: 50 klbf/in2, minimum.

Engineering note: This cable is useful in general, high temperature applications. (Used in 1 MHz multiplex applications in accordance with MIL-STD-1553.)

REQUIREMENTS:

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical inspection: Applicable

Diameter: Applicable.

Out-of-roundness: Applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:
Inner conductor to core: 1 pound, minimum; 6 pounds, maximum.

ELECTRICAL AND MECHANICAL:

Continuity: Applicable.

Spark test: 2,000 Vrms +10 percent, -0 percent.

Voltage withstanding: 1,000 V rms, +10 percent, -0 percent

Insulation resistance: 5,000 megohms per 1,000 feet.

Corona extinction voltage: Not applicable.

Characteristic impedance: 77 ohms ± 7 at 1 MHz.

RF transmission loss (attenuation): 1.4 dB per 100 feet, maximum at 1 MHz.

Standing wave ratio (return loss): Not applicable.

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Capacitance: 24 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: 5 percent, maximum.

Transmission unbalance: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.
Aging stability: Not applicable.

Stress-crack resistance: 230°C ± 5°C. Mandrel size is 5 times the jacket diameter for M17/176-00002.
150°C ± 5°C. Mandrel size is 10 times the jacket diameter for M17/176-00003.

Outer conductor integrity: Not applicable.

Cold bend: -55°C ± 2°C.

Dimensional stability: +200°C ± 5°C for M17/176-00002.
+150°C ± 5°C for M17/176-00003.

Inner conductor from core: .125 inch, maximum.
Inner conductor from jacket: .187 inch, maximum.

Temperature stabilization: Not applicable.
Contamination: Not applicable.
Bendability: Not applicable.
Flammability: Not applicable.

Weight: 18 pounds per 1,000 feet, maximum for M17/176-00002.
16 pounds per 1,000 feet, maximum for M17/176-00003.

Inductance: 118 nH per foot, minimum.

Measure inductance (L) at 1 MHz of a 10 foot 2 inch test cable using any suitable test instrument. The shield shall be floated. The end of the cable shall be shorted. The impedance (Z) shall then be calculated using the following formula: Where inductance = L (in henrys) and capacitance = C (in farads): $Z = \sqrt{L/C}$

Marking: Applicable.

Workmanship: Applicable.

Part or Identifying Number (PIN): See table II.

Supersession data: See table II.

TABLE II. Cross-reference of part number.

PIN	Superseded PIN or type designation
M17/176-00002	M17/176-00001
M17/176-00003	8712812-1 1/

1/ Air Force Drawing 8712812

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TABLE III. Qualification and Group A, B, and C Table.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
I	In-process inspection:		4.5	Entire sample		
	Continuity	3.7.1	4.8.2	Entire sample		
	Spark test	3.7.2	4.8.3	Entire sample		
	Voltage withstanding	3.7.3	4.8.4	Entire sample		
	Insulation resistance	3.7.4	4.8.5	Entire sample		
	Visual and mechanical inspection	3.6	4.8.1	Entire sample	Per Table VIII	A
	Diameter measurements.	3.6.1	4.8.1.1	Entire sample	Per Table VIII	A
	Out-of-roundness of jacket measurements	3.6.2	4.8.1.2	Entire sample	Per Table VIII	A
	Eccentricity of inner conductor	3.6.3	4.8.1.3	2	Per Table VIII	A
	Adhesion of conductors	3.6.4	4.8.1.4	4	Per Table VIII	A
	Physical dimensions	3.6	4.8.1	Entire sample	Per Table VIII	A
	Marking	3.8	4.8.1	Entire sample	Per Table VIII	A
	Workmanship	3.10	4.8.1	Entire sample	Per Table VIII	A
II	Corona extinction voltage	N/A				
	Characteristic impedance	3.7.6	4.8.7	1	Per Table VIII	A
	RF transmission loss (attenuation)	N/A				
	Standing wave ratio (return loss)	3.7.8	4.8.9	2	Per Table VIII	A
	Capacitance	3.7.9	4.8.10	1	Per Table VIII	B
	Capacitance Stability	N/A				
	Capacitance unbalance	3.7.11	4.8.12	1	Per Table VIII	B
	Transmission unbalance	N/A				
	Mechanically induced noise voltage	N/A				
	Time delay	N/A				
	Aging stability	N/A				
	Stress-crack resistance	3.7.16	4.8.17	4	Per Table VIII	C

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TABLE III. Qualification and Group A, B, and C Table – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
	Outer conductor integrity	N/A				
	Cold bend	3.7.18	4.8.19	3	Per Table VIII	B
	Dimensional stability	3.7.19	4.8.20	1	Per Table VIII	C
	Temperature stabilization	N/A				
	Contamination	N/A				
	Bendability	N/A				
	Flammability	N/A				
	Flame propagation	N/A				
	Acid gas generation	N/A				
	Halogen content	N/A				
	Immersion tests	N/A				
	Smoke index	N/A				
	Toxicity Index	N/A				
	Durometer hardness	N/A				
	Weathering	N/A				
	Abrasion resistance	N/A				
	Tear strength	N/A				
	Heat distortion	N/A				
	Physicals (aged)	N/A				
	Tensile strength and elongation	N/A				
	Inductance	3.7.36	4.8.38	1	Per Table VIII	B
	Weight	3.9	4.8.24	1	Per Table VIII	B

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-STD-1553 ASTM B298 ASTM B624

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2023-032)

Review activities:
Army – AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19,

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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